Class 21: Fixed Income I

April 16, 2026

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Today's Question

UK pension funds nearly collapsed in 2022 because of... bonds? How does that happen?

Positioning

Where We've Been	Where We Are	Where We're Going
Alternatives (equity-focused)	Bonds: pricing, yields, duration	Yield curve and interest rate risk

“UK Pensions Got Margin Calls”

Key themes: leveraged bond positions, liability-driven investing, and how “safe” assets can blow up

Learning Objectives

By the end of today's class, you should be able to:

1. Calculate bond prices from yields (and vice versa)
2. Define and calculate duration
3. Explain how leveraged bond positions can blow up

Zero Coupon Bonds

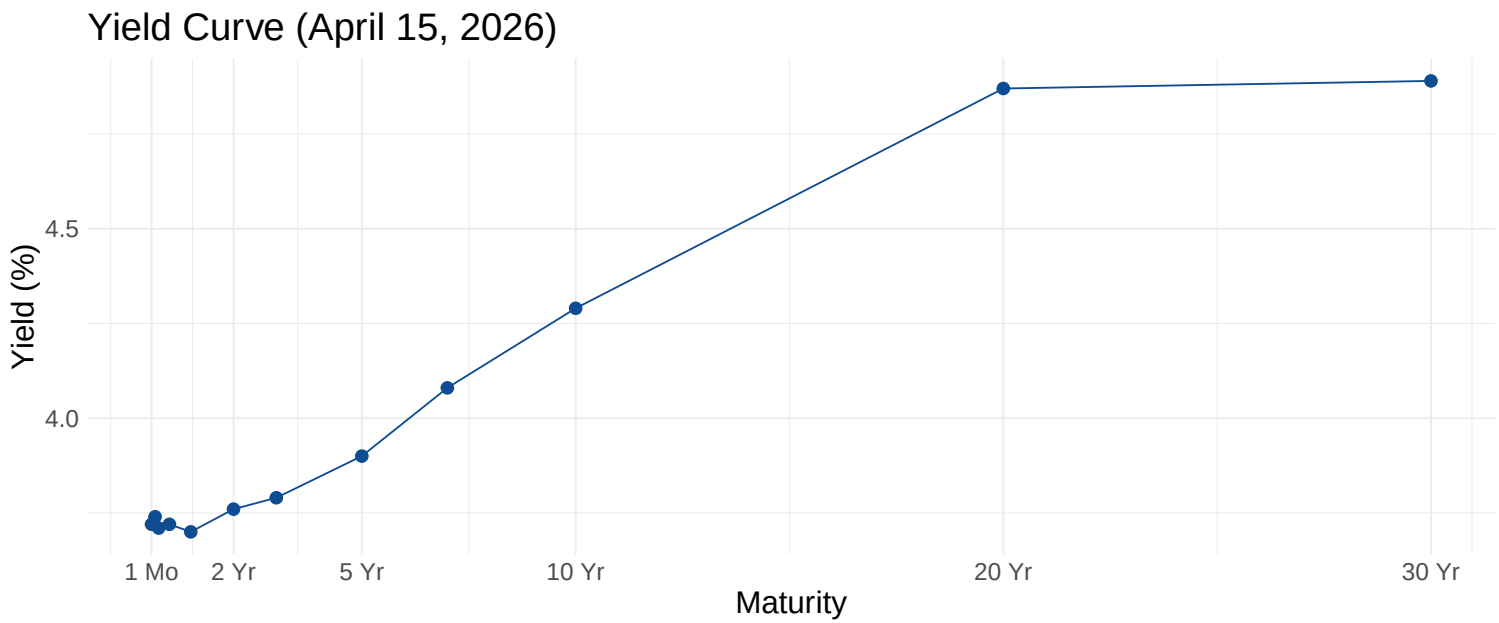
Consider a *risk-free* **zero-coupon** bond

- Face value (final payoff) of F with a maturity of T years
- If the interest rate for maturity T is r_T, then the current price of this bond is:

$$P_0 = \frac{F}{(1 + r_T)^T}$$

Yield Curve

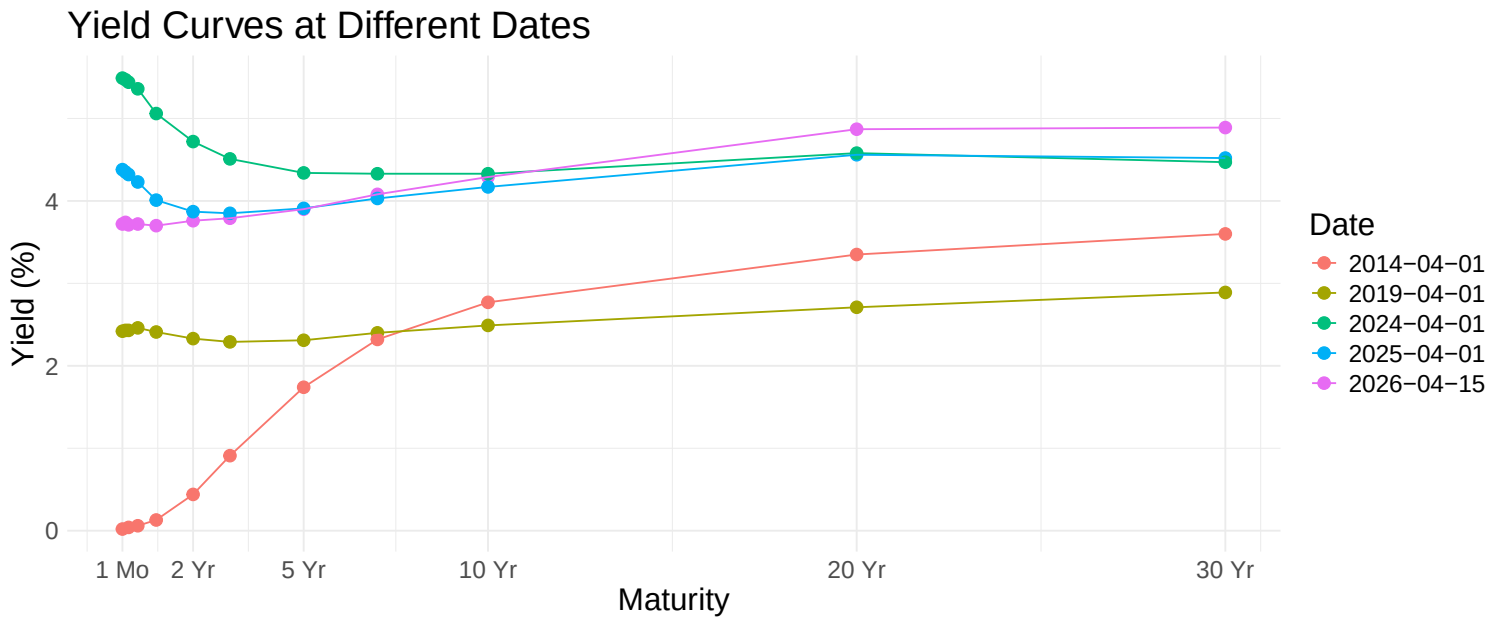
- Notation r_T implies variation in r over time
- **Yield curve** plots the term structure:



- In theory, produced from zero-coupon bonds or stripped Treasuries

Yield Curves Over Time

- The yield curve changes shape over time:



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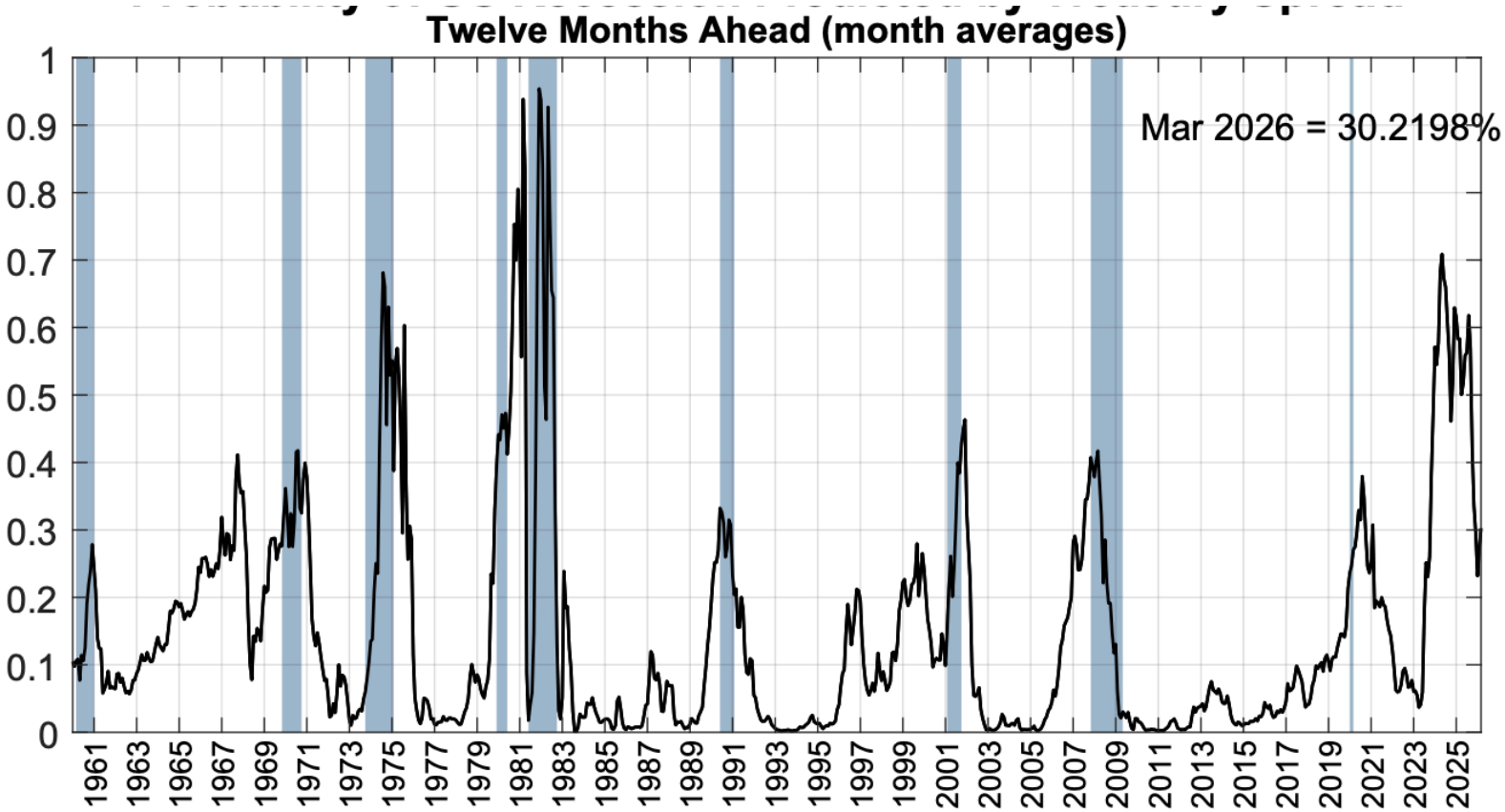
Yield Curve



*Parameters estimated using data from January 1959 to December 2009, recession probabilities predicted using data through Mar 2025.
The parameter estimates are α=-0.5333, β=-0.6330.

Updated 04-Apr-2025

Coupon Bonds

Coupon bonds pay coupon C in intervening periods (assume annually):

$$P_0 = \sum_{t=1}^{T} \frac{C}{(1+r_t)^t} + \frac{F}{(1+r_T)^T}$$

- C: interest or coupon payments
- T: number of periods (years) to maturity
- r_t: prevailing discount rate for horizon t

Yield to Maturity

- Suppose we observe coupon C, face value F and price P of a single bond
- Yield-to-maturity (YTM) solves for the discount rate that makes this price "right":

$$P = \sum_{t=1}^{T} \frac{C}{(1+\text{YTM})^t} + \frac{F}{(1+\text{YTM})^T}$$

- YTM is a weighted average of r_1 through r_T
- Plots of YTM by horizon approximate the true zero coupon yield curve

Yield to Maturity — Example

Consider a 10-year bond with face value of 100 and coupon payments of \$5/year

If these bonds trade **below** par, YTM exceeds coupon rate:

$$92.64 = \sum_{t=1}^{10} \frac{5}{(1+0.06)^t} + \frac{100}{(1+0.06)^{10}}$$

If bond trades at a **premium**, coupon rate exceeds discount rate:

$$108.11 = \sum_{t=1}^{10} \frac{5}{(1+0.04)^t} + \frac{100}{(1+0.04)^{10}}$$

Coupon Bonds and Yield Curves

- YTM collapses r_1 through r_T to a singular discount rate
- Suppose we want to solve for the entire yield curve
- If we want to know T interest rates over the yield curve, we need T bonds (roughly)
  - T equations and T unknowns
- Once we have interest rates for entire yield curve, can price any risk-free bond

Coupon Bonds and Yield Curves — Example

Use three bonds to solve for r_1, r_2, and r_3:

$$P_1 = \frac{C_1 + 100}{1 + r_1}$$

$$P_2 = \frac{C_2}{1 + r_1} + \frac{C_2 + 100}{(1 + r_2)^2}$$

$$P_3 = \frac{C_3}{1 + r_1} + \frac{C_3}{(1 + r_2)^2} + \frac{C_3 + 100}{(1 + r_3)^3}$$

Forward Rates

- Consider a contract in which we agree to lend money for 1 year, beginning 1 year in the future (repayment 2 years in the future)
  - $f_{1,2}$ is the *forward* rate for a 1-year loan issued 1 year from today
- Generally, $f_{j,k}$ is the forward rate from period j to k
  - $f_{0,1}$ is the normal 1-year spot rate, r_1
- A comment on notation: we use r_3 to mean today's prevailing interest rate for a 3-year horizon. This is different from BKM (in which r_3 refers to the 1-year rate which will prevail three years from now)

Forward Rates

Now consider two *equivalent* strategies:

1. Buy \$1 worth of a two-year zero coupon bond
   - Receive $(1 + r_2)^2$
2. Buy \$1 worth of a one-year zero coupon bond and a forward contract for 1 year beginning 1 year in the future
   - Receive $(1 + r_1)(1 + f_{1,2})$

By no arbitrage, since the bonds and contracts are riskless:

$$(1 + r_1)(1 + f_{1,2}) = (1 + r_2)^2$$

This implies that the 1- and 2-year bond rates set the forward rate:

$$f_{1,2} = \frac{(1 + r_2)^2}{1 + r_1} - 1$$

Next Class: April 21

- Topics: Fixed Income II
  - Forward rates and term structure
  - Risk aversion and the yield curve
  - Bond pricing relationships

- Matt Levine reading: “Startup Bank Had a Startup Bank Run”